

What's New In AOI

NEW FEATURES	DESCRIPTION	KEY BENEFITS
AI-powered defect classification	Uses machine learning to automatically classify defects and reduce false calls (e.g., missing, tombstoning, bridging)	Increases inspection accuracy by reducing false calls and escapes through intelligent defect analysis, continuously improving with data
Pure real 3D AOI	Full Z-axis measurement of every component/solder joint—not enhanced 2D—for highly accurate inspection	Offers precise height measurements of components and solder joints, ensuring better detection of lifted leads, coplanarity issues, and solder volume
MRS (multi-reflection suppression)	Eliminates reflections from shiny surfaces (solder, metal, ceramic) for true 3D measurement accuracy	Eliminates false reflections from shiny surfaces, improving detection accuracy on complex and reflective components
Closed-loop SPI-AOI integration	AOI and SPI systems communicate in real time to correct process errors upstream	Enables feedback between solder paste inspection (SPI) and AOI systems, allowing real-time process correction and yield improvement
Deep learning vision systems	AI engines learn from historical defect data to improve inspection accuracy over time	Enhances capability to recognise complex patterns and borderline defects, significantly reducing programming effort and human intervention
Real-time inline X-ray + AOI combo	AOI systems with inline 3D CT X-ray for hidden solder inspection in one platform	Combines surface and hidden defect detection (e.g., BGA voids, internal shorts) in a single inspection pass, improving coverage and throughput
Remote and predictive maintenance (IoT)	AOI machines send status alerts, predict failures, and optimise uptime using IIoT/Industry 4.0	Minimises machine downtime through proactive alerts and remote diagnostics, improving machine uptime and operational efficiency
Multi-camera 360° inspection	Use of multiple side cameras (up to 9) for component tilt/lead inspection from all angles	Provides side-view and angled inspections, ensuring detection of side defects like lifted pins, label misplacement, and tall component alignment
Ultra-high resolution imaging (25MP+)	Ultra-HD image capture improves accuracy on 01005, micro-BGA, and fine-pitch components	Captures finer defects and micro-components on high-density PCBs, improving resolution, and precision in inspection
Automatic program generation	AI-based auto-programming minimises setup time for new PCBs	Reduces setup time and operator skill dependency by auto-generating inspection programs using CAD/Gerber and part libraries

PROSEM

Website: prosem.in
Address: No.51/1/8, Site-4, Sahibabad Industrial Area, Sahibabad, Ghaziabad, Uttar Pradesh -201010
Email: sales@prosem.in
Phone: 0120-4284733
Brand(s) Supplied: Saki, ZhenHuaXing (VCTA)

ROCINANTE TECHNOLOGIES

Website: rocinantetechnologies.com
Address: 80 A, Udyog Vihar, Phase V, Gurgaon- 122016, Haryana
Email: info@rocinantetechnologies.com
Phone: 8860829161
Brand(s) Supplied: Seamark

SMT TECH LINK

Website: www.smttechlink.in
Address: D Wing-304, Anand Homes, Adharwadi Road, Near Pawan Dham, Ganpati Chowk, Kalyan, Thane, Maharashtra -421301
Email: sales@smttechlink.in

Phone: 08047769721
Brand(s) Supplied: Maker ray, Persci

TESSCORN NANOSCIENCE INC.

Website: www.tesscorn-nanoscience.com
Address: 50, 3rd floor, 100 Feet Road, Koramangala 2nd Block, Bengaluru, Karnataka -560034
Email: info@tesscorn-nanoscience.com
Phone: 080 25729425
Brand(s) Supplied: Faroad

TEST & RESEARCH INDIA PVT LTD

Website: testindia.co.in
Address: Unit Nol 947, 9th Floor, JMD Megapolis, Sector - 48, Sohna Road, Gurgaon, Haryana -122018
Email: ravi@testindia.co.in
Number: 9818143339
Brand(s) Supplied: TRI

TRANSTECHNOLOGY INDIA PVT LTD

Website: www.trans-tec.com
Address: 601A, 6th Floor, Tower-2 DLF Corporate Greens, Sector-74A, Gurgaon,

Haryana -122004
Email: trans-tec.india@trans-tec.com
Phone: 9845485937
Brand(s) Supplied: Yamaha

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